

Pitch to Postgres: Designing and Querying a Relational Database for FIFA World Cup Data

DS604 Introduction to Data Management
Milestone 2: Schema

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Data: The Fjelstul World Cup Database by Joshua C. Fjelstul (CC-BY-SA 4.0)

1. Summary

We designed a normalised PostgreSQL schema for all 27 tables of the Fjelstul World Cup Database, plus one small lookup table (`positions`), giving **28 tables**. The DDL is in `sql/ddl.sql` (reproduced in full in Section 7).

Item	Count
Tables	28 (27 from the dataset + 1 lookup)
Columns	163 kept from 405 source columns, plus 2 in <code>positions</code>
Primary keys	28 (13 of them composite)
Foreign keys	49 (36 of them composite)
UNIQUE constraints	14
CHECK constraints	63
NOT NULL columns	147 of 165
Secondary indexes	21

How it was tested (local PostgreSQL 18):

1. **Runs repeatably.** `ddl.sql` runs without errors, and running it again changes nothing. It contains no DROP, TRUNCATE or DELETE, so it is safe on the course server.
2. **Accepts the real data.** All 27 CSV files were trial-loaded with `pandas.to_sql(if_exists="append")`, the method required for Milestone 3. All **57,269 rows** were accepted under every key and constraint.
3. **Rejects bad data.** 14 deliberately invalid changes were each rejected by the intended constraint (Section 6).
4. **Loses no information.** All 33 checks pass, showing that every dropped column can be recomputed exactly from the columns we keep (Section 3.2).

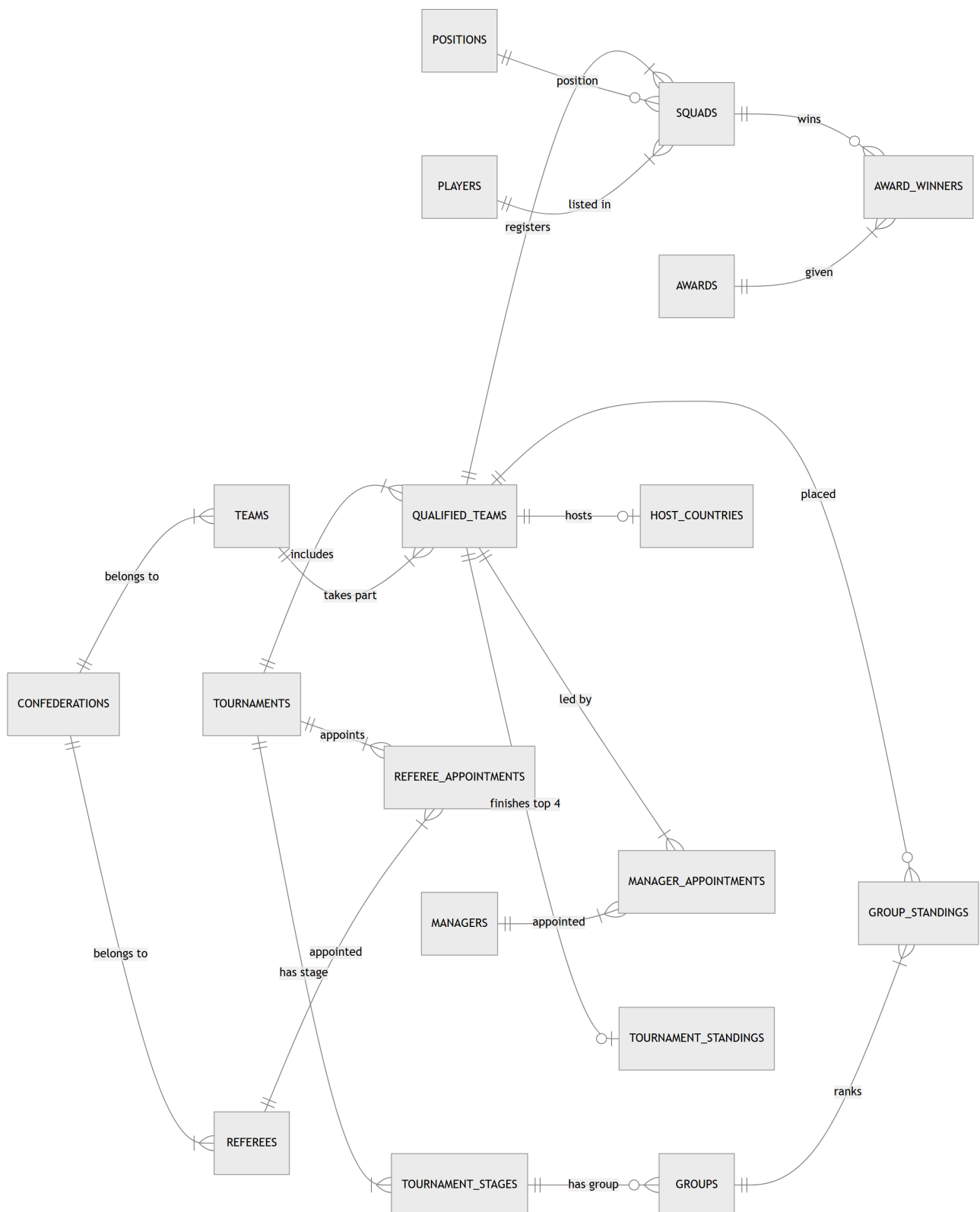
The design work also exposed **one more inconsistency in the source**. For the 1982 second group stage, `matches` names the groups "Group 1"–"Group 4", while `groups` and `group_standings` call them "Group A"–"Group D". The foreign key from `matches` to `groups` caught it on the first trial load. The loader matches each group by its identical set of three teams (1=A, 2=B, 3=C, 4=D), and this finding has been added to the Milestone 0 report. *(Added during Milestone 3: an integrity query also found that the six 1950 final-round matches have no group name, while `groups` records that round as "Group 1". The loader assigns them to that group by their teams.)*

2. ER diagram

The conceptual ER diagram is split into two sheets for readability; entities that link the two themes appear on both. Crow's-foot ends show cardinality. A circle means zero is possible ("optional"), and a bar means at least one ("mandatory"). Cardinalities were **measured on the loaded data**, not assumed. Both diagrams are generated from the live database catalog by `src/make_diagrams.py`.

ER diagram 1/2: tournaments, teams, people and awards

Pitch to Postgres: Designing and Querying a Relational Database for FIFA World Cup Data | crow's foot notation, cardinalities measured on the loaded data | data: Fjelstul World Cup Database (CC-BY-SA 4.0)



ER diagram 2/2: matches and match events

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3. Design explanation

3.1 Normal form

The schema is in **third normal form (3NF)**, with one deliberate and controlled exception (Section 3.5).

- **1NF:** every column holds a single value. The source column `players.list_tournaments` held a comma-separated list such as "1998, 2002, 2006", so it was removed; the same information comes from `squads`.
- **2NF:** no column depends on only part of a composite key. In the source, for example, `team_name` in `squads` depends on `team_id` alone, not on the whole key (`tournament_id`, `team_id`, `player_id`), so copied names like this were removed.
- **3NF:** no column depends on another non-key column. Examples removed: `teams.confederation_name` depends on `confederation_id`; `matches.home_team_score_margin` depends on the two scores; `bookings.sending_off` depends on `red_card` and `second_yellow_card`.

3.2 What was removed, and proof that nothing is lost

Three kinds of column were removed:

- **Copies** of attributes that belong to another table (names, codes, dates, stadium location, confederation). These cause update anomalies: renaming a team would mean updating 16 tables.
- **Same-row derived values**, such as score text, margins, result labels and win/draw flags, `minute_label`, `match_period`, `going_off`, `substitute`, `sending_off`, `played`, `goal_difference` and `points`.
- **Summary values derivable from other tables**, such as player position flags, `count_tournaments`, tournament format flags, the `winner` and `host_country` text, match and team counts, `unbalanced_groups` and `shared`.

`key_id` is only the row number of each file, so it was not kept.

Before dropping anything we proved each column redundant on the actual data.

`clean.verify_dropped_columns()` runs **33 checks, and all pass**. For example, `result` is reproduced exactly from the two scores plus the shoot-out scores, and `points` is reproduced exactly from wins and draws with 2 points per win up to 1990 and 3 from 1994. Everything removed can be rebuilt with a query, and Milestone 5 provides views that do so (for example, a match results view with result labels).

Table	Source columns	Kept	Dropped columns
<code>confederations</code>	5	4	<code>key_id</code>
<code>awards</code>	5	4	<code>key_id</code>
<code>stadiums</code>	8	7	<code>key_id</code>
<code>teams</code>	11	8	<code>key_id</code> , <code>confederation_name</code> , <code>confederation_code</code>
<code>players</code>	12	5	<code>key_id</code> , <code>goal_keeper</code> , <code>defender</code> , <code>midfielder</code> , <code>forward</code> , <code>count_tournaments</code> , <code>list_tournaments</code>
<code>managers</code>	6	5	<code>key_id</code>
<code>referees</code>	9	6	<code>key_id</code> , <code>confederation_name</code> , <code>confederation_code</code>
<code>tournaments</code>	18	5	<code>key_id</code> , <code>host_country</code> , <code>winner</code> , <code>host_won</code> , <code>count_teams</code> , <code>group_stage</code> , <code>second_group_stage</code> , <code>final_round</code> , <code>round_of_16</code> , <code>quarter_finals</code> , <code>semi_finals</code> , <code>third_place_match</code> , <code>final</code>

Table	Source columns	Kept	Dropped columns
tournament_stages	16	9	key_id, tournament_name, group_stage, knockout_stage, unbalanced_groups, count_matches, count_replays
groups	7	3	key_id, tournament_name, stage_name, count_teams
qualified_teams	8	3	key_id, tournament_name, team_name, team_code, count_matches
host_countries	7	2	key_id, tournament_name, team_name, team_code, performance
tournament_standings	7	3	key_id, tournament_name, team_name, team_code
group_standings	19	11	key_id, tournament_name, stage_name, team_name, team_code, played, goal_difference, points
squads	12	5	key_id, tournament_name, team_name, team_code, family_name, given_name, position_name
manager_appointments	10	3	key_id, tournament_name, team_name, team_code, family_name, given_name, country_name
referee_appointments	10	2	key_id, tournament_name, family_name, given_name, country_name, confederation_id, confederation_name, confederation_code
matches	37	17	key_id, tournament_name, match_name, stage_name*, group_stage, knockout_stage, stadium_name, city_name, country_name, home_team_name, home_team_code, away_team_name, away_team_code, score, home_team_score_margin, away_team_score_margin, score_penalties, result, home_team_win, away_team_win, draw
team_appearances	36	9	key_id, tournament_name, match_name, stage_name, group_name, group_stage, knockout_stage, replayed, replay, match_date, match_time, stadium_id, stadium_name, city_name, country_name, team_name, team_code, opponent_name, opponent_code, away_team, goal_differential, extra_time, penalty_shootout, result, win, lose, draw
referee_appearances	15	3	key_id, tournament_name, match_name, match_date, stage_name, group_name, family_name, given_name, country_name, confederation_id, confederation_name, confederation_code
manager_appearances	17	4	key_id, tournament_name, match_name, match_date, stage_name, group_name, team_name, team_code, home_team, away_team, family_name, given_name, country_name
player_appearances	22	7	key_id, tournament_name, match_name, match_date, stage_name, group_name, team_name, team_code, home_team, away_team, family_name, given_name, shirt_number, position_name, substitute
goals	27	10	key_id, tournament_name, match_name, match_date, stage_name, group_name, team_name, team_code, home_team, away_team, family_name, given_name, shirt_number, player_team_name, player_team_code, minute_label, match_period
bookings	26	10	key_id, tournament_name, match_name, match_date, stage_name, group_name, team_name, team_code, home_team, away_team, family_name, given_name, shirt_number, minute_label, match_period, sending_off
substitutions	24	8	key_id, tournament_name, match_name, match_date, stage_name, group_name, team_name, team_code, home_team, away_team, family_name, given_name, shirt_number, minute_label, match_period, going_off
penalty_kicks	19	6	key_id, tournament_name, match_name, match_date, stage_name, group_name, team_name, team_code, home_team, away_team, family_name, given_name, shirt_number
award_winners	12	4	key_id, tournament_name, award_name, shared, family_name, given_name, team_name, team_code

* `matches.stage_name` is replaced by `stage_number` (Section 3.6).

3.3 Natural keys instead of surrogate keys

Every table keeps the source's own identifiers (`team_id` such as T-03, `player_id` such as P-04418, `match_id` such as M-1930-01) as its primary key. We did **not** add surrogate keys (auto-numbered integers).

- The ids are already **unique, never missing and stable**, which we verified for every table in Milestone 0.
- The course rule is to load each CSV with `to_sql(if_exists="append")`. With natural keys, child tables can be appended directly. With surrogate keys, every foreign key in the 20 child tables would have to be looked up and replaced during loading, which adds work and room for error.
- The ids carry meaning that helps when checking results (WC-1930, M-1930-01), and CHECK constraints enforce their format, for example `team_id ~ '^T-[0-9]{2}$'`. The match id must also contain the tournament year.
- `key_id` is **not** used as a key. It is only the row position in each file and identifies nothing in the real world.

3.4 Composite keys

Bridge and event tables use composite primary keys made of the foreign keys that identify one fact:

Table	Primary key	Why
<code>qualified_teams</code>	(tournament_id, team_id)	A team takes part in a tournament at most once.
<code>squads</code>	(tournament_id, team_id, player_id)	One squad entry per player per team per tournament. A further UNIQUE (tournament_id, player_id) enforces that a player is in only one squad per tournament (true for all 10,142 rows).
<code>manager_appointments</code>	(tournament_id, team_id, manager_id)	Joint managers exist (12 team-tournaments), so <code>manager_id</code> must be part of the key.
<code>manager_appearances</code>	(match_id, team_id, manager_id)	Same reason; 42 team-matches had two managers.
<code>team_appearances</code>	(match_id, team_id)	Two rows per match, one per team.
<code>player_appearances</code>	(match_id, player_id)	A player appears at most once in a match.
<code>groups</code>	(tournament_id, stage_number, group_name)	Group names can repeat across stages (in 1950 "Group 1" exists in both the first round and the final round), so (tournament_id, group_name) is not unique.
<code>group_standings</code>	(tournament_id, stage_number, group_name, position)	One team per table position. UNIQUE (tournament_id, stage_number, team_id) adds that a team is in only one group per stage.
<code>tournament_standings</code>	(tournament_id, position)	Positions 1–4 are unique; UNIQUE (tournament_id, team_id) as well. There are no shared places in the data.
<code>award_winners</code>	(tournament_id, award_id, player_id)	Ties produce several winners (the 1962 Golden Boot had six).

3.5 Controlled redundancy: tournament_id in event tables

Strictly, `tournament_id` in `goals` (and the other event tables) depends on `match_id`, so keeping it is a transitive dependency. We keep it on purpose, because it makes a much stronger integrity rule possible:

- The foreign key (tournament_id, player_team_id, player_id) -> `squads` guarantees that **every goal scorer was in their team's squad for that tournament**. The same rule applies to cards, substitutions,

shoot-out kicks, appearances and awards. Without `tournament_id` this rule cannot be written as a foreign key.

- To make sure the extra column **can never disagree** with the match, each event table also has the foreign key `(match_id, tournament_id) -> matches(match_id, tournament_id)`. It points at a UNIQUE constraint on `matches`. The redundancy is therefore enforced by the database and cannot cause an update anomaly.

`team_appearances` is likewise a derived, team-centred copy of `matches`, kept because it makes per-team queries much simpler. Its consistency is enforced in two ways:

- **Opponent check:** a self-referencing foreign key, `(match_id, opponent_id) -> team_appearances(match_id, team_id)`, ensures the opponent has its own row in the same match. It is declared `DEFERRABLE INITIALLY DEFERRED`, so the two rows of a match can be inserted in either order and are checked at COMMIT.
- **Team check:** a foreign key ensures each team qualified for that tournament.

3.6 Stages and groups

The source stores stage names inconsistently. `groups` and `group_standings` say "first round" (1950) and "first group stage" (1974–1982) where `tournament_stages` and `matches` say "group stage". We therefore:

- store `stage_name` **only** in `tournament_stages`, with UNIQUE (`tournament_id, stage_name`) and a CHECK restricting it to the 8 stage names in the data;
- link `groups`, `group_standings` **and** `matches` to stages through `stage_number`. During loading, `matches.stage_number` is looked up from `tournament_stages`, since the source `matches` has only the stage name;
- give `matches` a foreign key `(tournament_id, stage_number, group_name) -> groups`. Knockout matches have `group_name` NULL, so the key is not checked for them (standard MATCH SIMPLE behaviour). This foreign key detected the 1982 "Group 1–4" / "Group A–D" inconsistency.

3.7 Historical teams

West Germany, East Germany, the Soviet Union, Yugoslavia, Serbia and Montenegro, Czechoslovakia, Zaire and the Dutch East Indies remain **separate rows in teams, with their source ids**, because that is how they played. Consequences:

- `team_code` is **NOT UNIQUE**: Germany and West Germany share DEU. It has NOT NULL and a three-capital-letters CHECK, but it is not a key; `team_id` is.
- We did **not** add a "lineage" table merging teams (for example West Germany into Germany). FIFA's credit to successor federations is outside knowledge, not something in the dataset. If a query combines teams, it says so explicitly.
- `teams.confederation_id` is the **current** confederation (Australia: AFC, although it played in 1974 and 2006 as an OFC member). This is documented in the table comment.

3.8 Missing values, types and the positions lookup

Missing values. These become SQL **NULL** instead of placeholder values: - the text "not available" or "not applicable"; - shirt number 0, which means unknown; - shoot-out scores of 0 or "0-0" for the 870 matches without a shoot-out. A CHECK enforces that shoot-out scores are present **if and only if** there was a shoot-out.

Types. - DATE for dates, TIME for kick-off times and BOOLEAN for all 0/1 flags. This enables date arithmetic (ages) and readable CHECKs. - SMALLINT for small counts and minutes, INTEGER for stadium capacity. - VARCHAR(n) sized with headroom above the longest observed value (for example, the longest family name is 23 characters and the column allows 60).

Positions lookup. `positions` is a **lookup table** we added. In the source, `position_name` depends on `position_code`, a 3NF violation repeated in two tables. The 21 code/name pairs are inserted by the DDL, exactly as they appear in the data. `squads.position_code` is further limited by CHECK to the four squad codes (GK, DF, MF, FW).

3.9 CHECK and UNIQUE constraints that encode football rules

Constraint	Rule
<code>ck_match_shootout</code>	A shoot-out implies extra time, a level score, and different, non-NULL shoot-out scores. Without a shoot-out, both shoot-out scores are NULL.
<code>ck_match_replayed_draw</code> , <code>ck_match_replay</code>	A replayed match was drawn, and a match cannot be both the original and the replay.
<code>ck_match_teams_differ</code> , <code>ck_ta_opponent</code>	A team cannot play itself.
<code>ck_goal_own_goal</code>	<code>own_goal</code> is TRUE exactly when the credited team differs from the scorer's team. Verified on all 2,548 goals.
<code>ck_goal_not_both</code>	A goal cannot be both an own goal and a penalty.
<code>ck_booking_some_card</code> , <code>ck_booking_second_yellow</code>	A booking shows at least one card, and a second yellow implies a yellow.
minute CHECKs	<code>minute_regulation</code> between 1 and 120; <code>minute_stoppage</code> ≥ 0.
<code>ck_tournament_id_year</code> , <code>ck_tournament_start_year</code> , <code>ck_match_id_year</code>	Ids agree with the tournament year.
<code>uq_squad_shirt</code>	Shirt numbers are unique within a squad (NULLs, meaning unknown, are allowed).
<code>uq_stadium_name_city</code>	Stadium names are unique within a city (there are two Olympiastadions, in Berlin and Munich).

Not enforced by the schema (they would need triggers, which are outside this course's scope): - a match date falls between its tournament's start and end dates; - `group_name` is present exactly for group-stage matches; - the goals in `goals` add up to the score in `matches`.

These are checked by queries instead; the last is query Q30 in Milestone 4.

3.10 Indexes

PostgreSQL automatically indexes every primary key and UNIQUE constraint. We added **21 secondary indexes** on foreign-key columns that the analytical queries join or filter on and that are *not* the first column of an existing index:

- `goals(player_id)`: top scorers (Q10, Q11).
- `goals(match_id, team_id)`, `bookings(match_id, team_id)`, `substitutions(match_id, team_id)`: per-match aggregation (Q9, Q22–Q25).
- `team_appearances(team_id, opponent_id)`: team records and head-to-head (Q2, Q7).
- `matches(stadium_id)`, `matches(tournament_id, stage_number)`, plus the home and away team columns: venues and stage analyses.
- `squads(player_id)`, `player_appearances(player_id)`, and the referee, manager and award lookups.

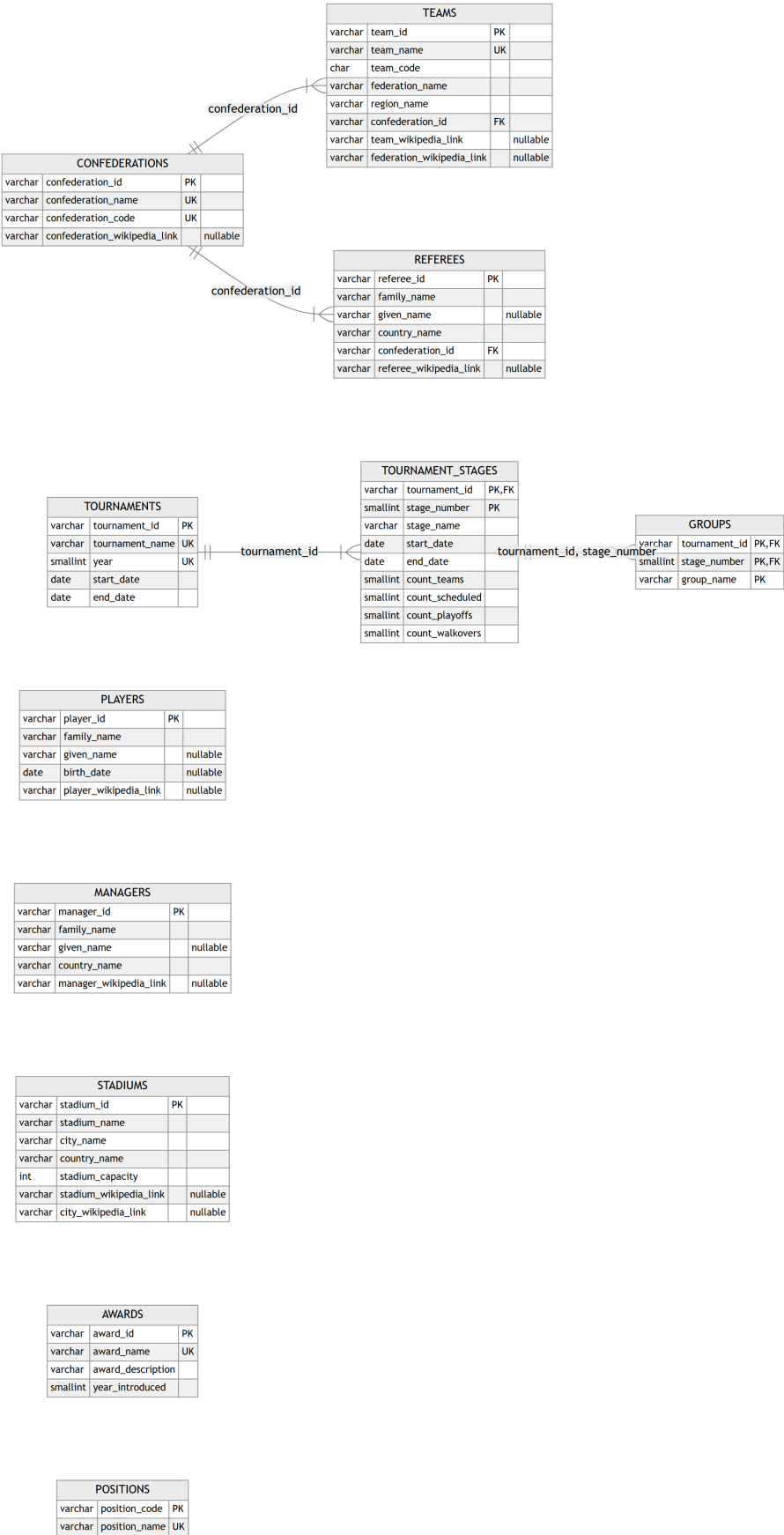
PostgreSQL does not index foreign keys automatically, and without these indexes it would also have to scan the child tables when a parent row is deleted or its key is updated. The dataset is small (57,269 rows), so the gain is modest, but the choices follow standard practice and the query plans in Milestone 4 can demonstrate them.

4. Relational schema diagrams

Five sheets, with columns, types, primary keys (PK), foreign keys (FK), single-column unique keys (UK) and nullable columns. Tables owned by another sheet appear only with their key columns and the columns referenced from that sheet. The labels on the lines are the foreign-key columns.

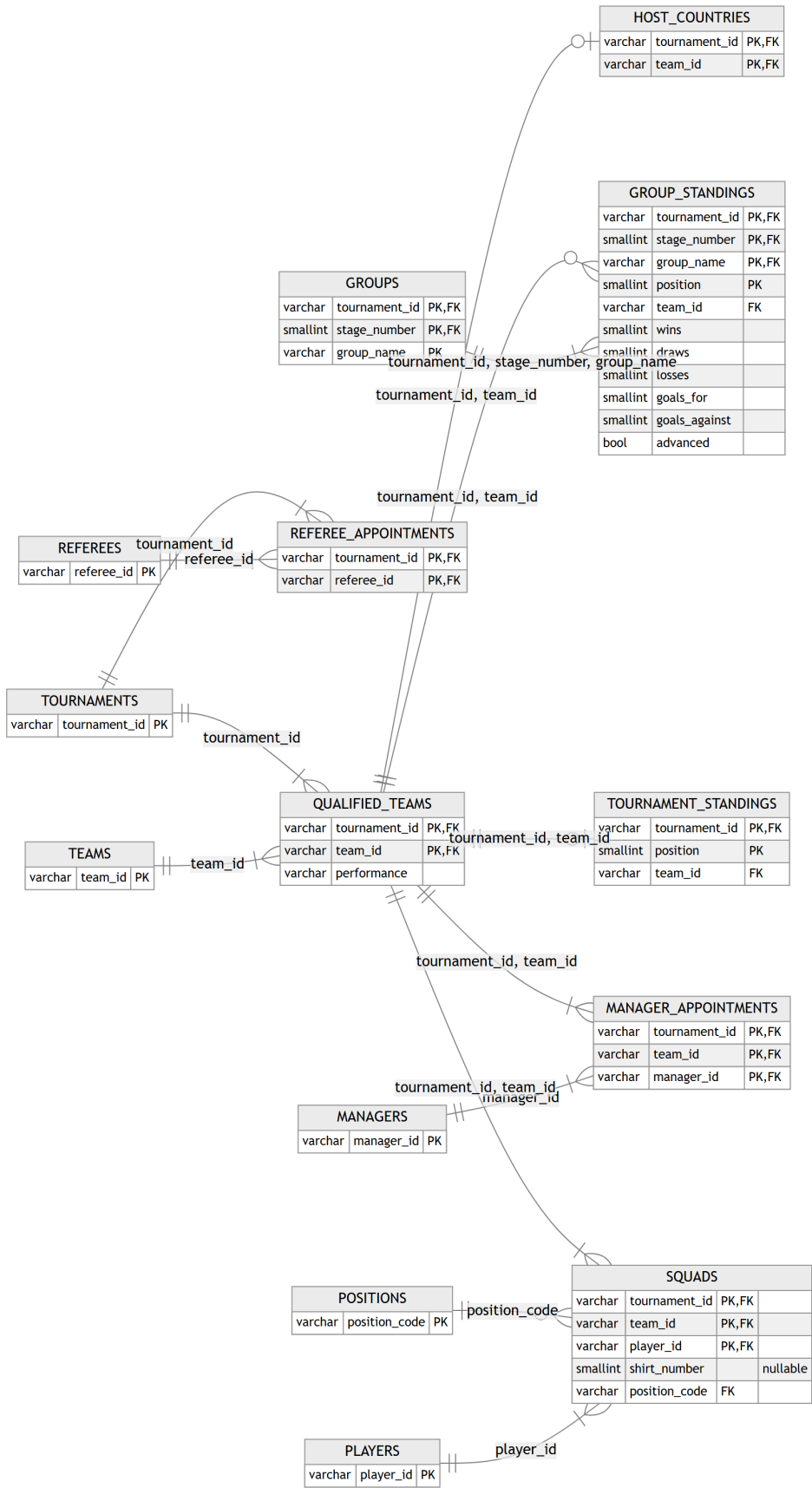
Relational schema 1/5: reference data, people and tournament structure

Pitch to Postgres: Designing and Querying a Relational Database for FIFA World Cup Data | crow's foot notation, cardinalities measured on the loaded data | tables from other sheets show key columns only | data: Fjelstul World Cup Database (CC-BY-SA 4.0)



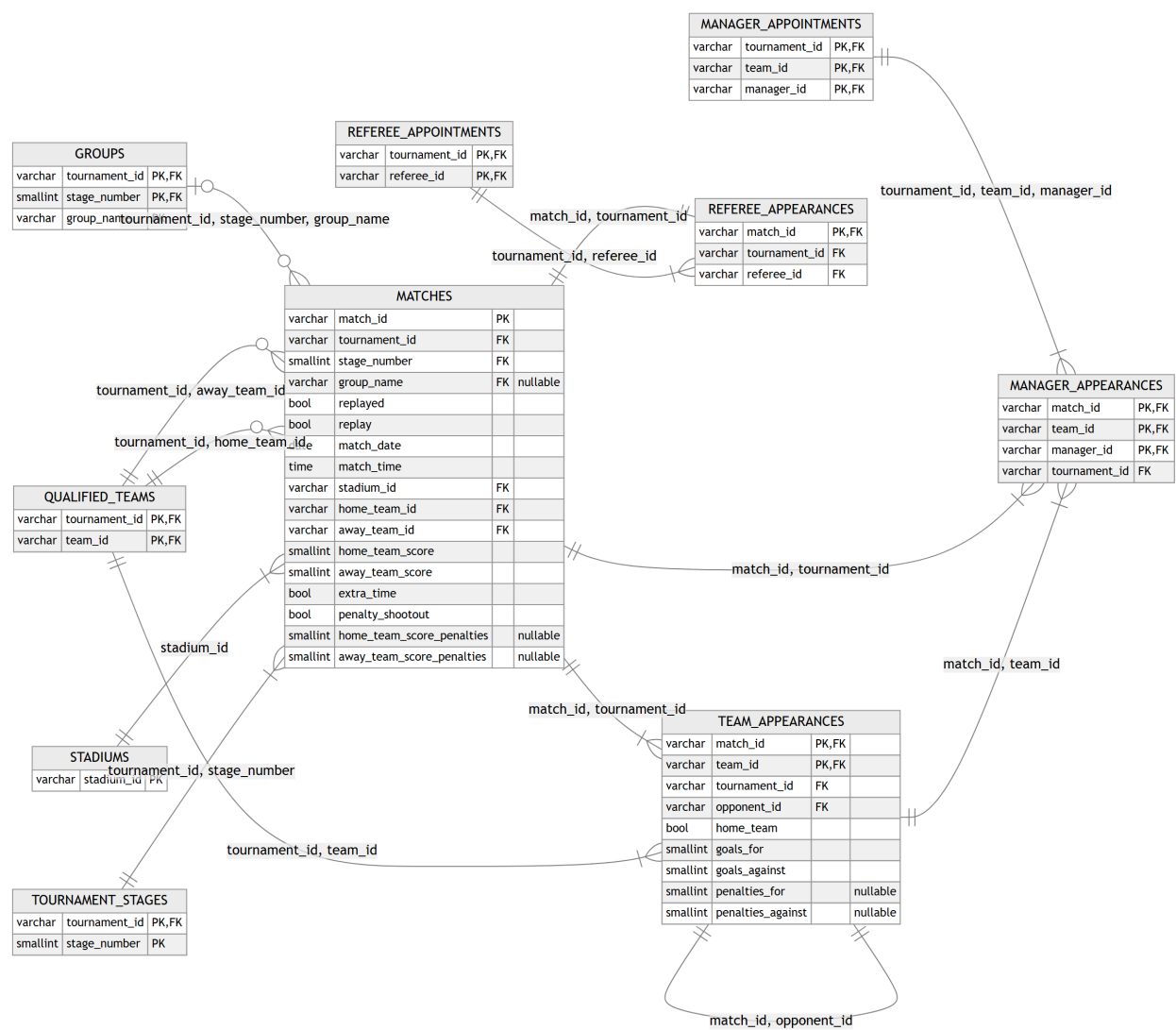
Relational schema 2/5: participation and standings

Pitch to Postgres: Designing and Querying a Relational Database for FIFA World Cup Data | crow's foot notation, cardinalities measured on the loaded data | tables from other sheets show key columns only | data: Fjellstul World Cup Database (CC-BY-SA 4.0)



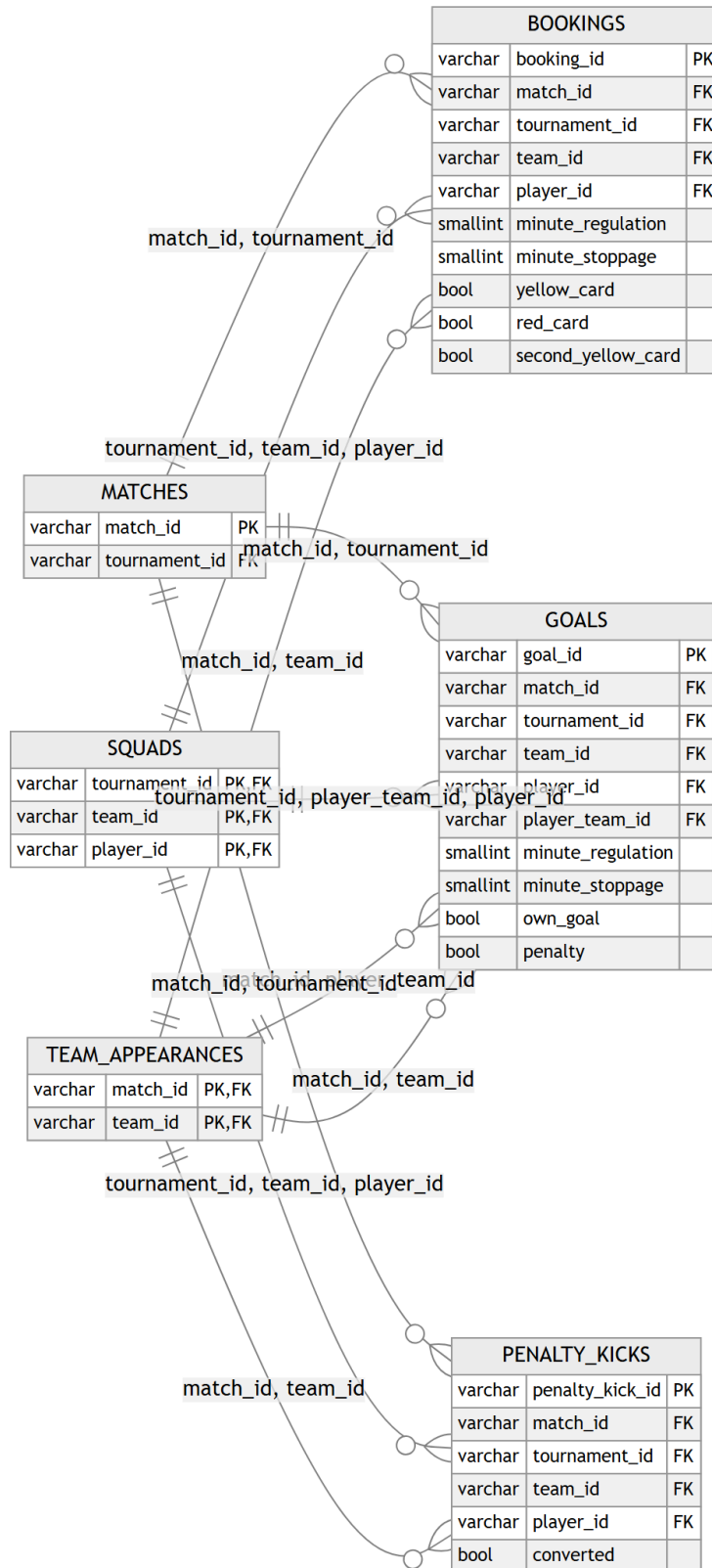
Relational schema 3/5: matches, team, referee and manager appearances

Pitch to Postgres: Designing and Querying a Relational Database for FIFA World Cup Data | crow's foot notation, cardinalities measured on the loaded data | tables from other sheets show key columns only | data: Fjølstul World Cup Database (CC-BY-SA 4.0)



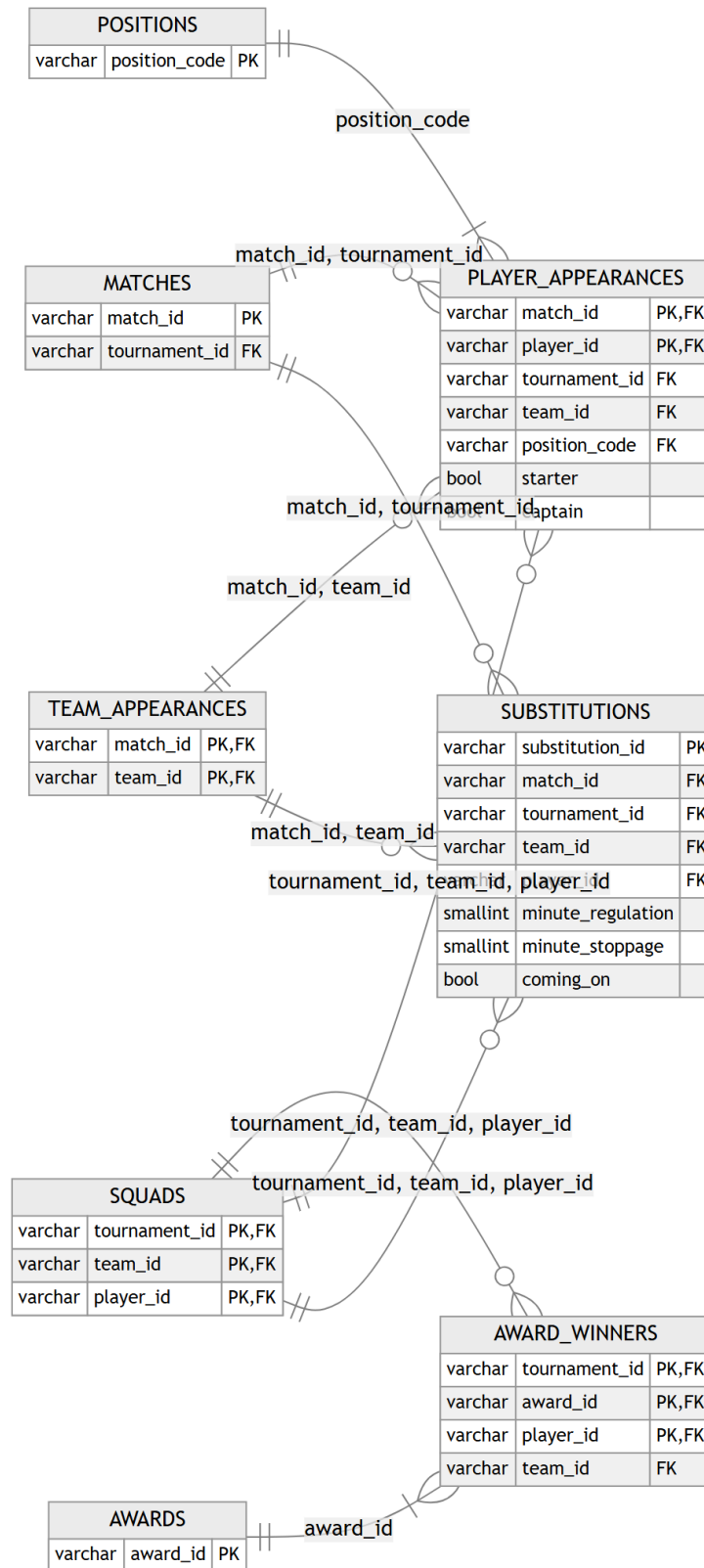
Relational schema 4/5: goals, cards and shoot-out kicks

Pitch to Postgres: Designing and Querying a Relational Database for FIFA World Cup Data | crow's foot notation, cardinalities measured on the loaded data | tables from other sheets show key columns only | data: Fjelstul World Cup Database (CC-BY-SA 4.0)



Relational schema 5/5: line-ups, substitutions and awards

Pitch to Postgres: Designing and Querying a Relational Database for FIFA World Cup Data | crow's foot notation, cardinalities measured on the loaded data | tables from other sheets show key columns only | data: Fjelstul World Cup Database (CC-BY-SA 4.0)



5. Table dependency order

`ddl.sql` creates the tables, and Milestone 3 loads them, in this order, so every parent exists before its children:

1. confederations, positions, awards, stadiums, teams
2. players, managers, referees
3. tournaments, tournament_stages, groups
4. qualified_teams, host_countries, tournament_standings, group_standings
5. squads, manager_appointments, referee_appointments
6. matches, team_appearances, referee_appearances, manager_appearances
7. player_appearances, goals, bookings, substitutions, penalty_kicks, award_winners

6. Testing on the local database

Test	Result
Run <code>ddl.sql</code> on an empty database	28 tables created
Run <code>ddl.sql</code> a second time	No errors, no changes (IF NOT EXISTS; the <code>positions</code> seed uses ON CONFLICT DO NOTHING)
Trial load of all 27 tables with <code>to_sql(if_exists="append")</code>	57,269 / 57,269 rows accepted; every row count matches its CSV
Redundancy proofs for dropped columns	33 / 33 pass
Negative tests (each inside a transaction that was rolled back)	14 / 14 rejected by the intended constraint

The 14 rejected changes:

1. An own-goal flag that contradicts the teams.
2. A match with the same team on both sides.
3. A shoot-out match with an unequal score.
4. Shoot-out scores on a match without a shoot-out.
5. A booking for a player who is not in that team's squad.
6. A duplicate shirt number in a squad.
7. A detailed position code (CB) in a squad.
8. A tournament id that disagrees with its year.
9. A match id that disagrees with its tournament.
10. A goal credited to a team that did not play in the match.
11. A team appearance without its opponent's row (rejected at COMMIT by the deferred foreign key).
12. Deleting a team that has matches.
13. A booking row with no card.
14. A goal in minute 130.

Tools: `src/apply_ddl.py` runs the DDL. Its `--reset-local` option drops everything first, and it refuses to do so unless the host is local. `src/make_diagrams.py` regenerates the diagrams from the database catalog.

7. Full DDL (`sql/ddl.sql`)

```
-- =====  
-- Pitch to Postgres: Designing and Querying a Relational Database for FIFA World Cup Data  
-- DS604 Introduction to Data Management - Milestone 2: Schema (DDL)
```

```

-- Group: Devanshi Dudhatra (202618027), Pujita Sunapu (202618003), Rahul Saha (202618037)
--
-- Data: Joshua C. Fjelstul, The Fjelstul World Cup Database,
--       https://github.com/jfjelstul/worldcup, CC-BY-SA 4.0.
--       Modifications: the source tables are normalised (copied columns, row numbers and
--       same-row derived columns are removed). Text markers such as 'not available' become NULL.
--       One lookup table (positions) is added. No data values are changed.
--
-- Target: PostgreSQL 12 or later. Standard SQL only, apart from the ~ regex operator in CHECKs.
-- Safe to run more than once: CREATE ... IF NOT EXISTS, and no DROP, TRUNCATE or DELETE.
-- Tables are created in dependency order (parents before children).
-- Tables are created in the current schema (set DB_SCHEMA in .env to choose one).
-- =====

-- -----
-- 1. Reference tables
-- -----

CREATE TABLE IF NOT EXISTS confederations (
    confederation_id      VARCHAR(5)    PRIMARY KEY CHECK (confederation_id ~ '^CF-[0-9]+$'),
    confederation_name     VARCHAR(100)  NOT NULL UNIQUE,
    confederation_code     VARCHAR(10)   NOT NULL UNIQUE,
    confederation_wikipedia_link VARCHAR(255)
);
COMMENT ON TABLE confederations IS 'One row per continental confederation (AFC, CAF, CONCACAF, CONMEBOL, OFC, UEFA).';

-- Lookup added by the group: position codes used in squads and player_appearances.
-- Values are exactly the code/name pairs that occur in the source files.
CREATE TABLE IF NOT EXISTS positions (
    position_code VARCHAR(3)  PRIMARY KEY,
    position_name VARCHAR(30) NOT NULL UNIQUE
);
COMMENT ON TABLE positions IS 'Lookup of playing positions (codes and names exactly as in the source files).';

INSERT INTO positions (position_code, position_name) VALUES
    ('GK', 'goal keeper'), ('DF', 'defender'), ('MF', 'midfielder'), ('FW', 'forward'),
    ('SW', 'sweeper'), ('CB', 'center back'), ('LB', 'left back'), ('RB', 'right back'),
    ('LWB', 'left wing back'), ('RWB', 'right wing back'), ('DM', 'defensive midfielder'),
    ('CM', 'center midfielder'), ('LM', 'left midfielder'), ('RM', 'right midfielder'),
    ('AM', 'attacking midfielder'), ('LW', 'left winger'), ('RW', 'right winger'),
    ('SS', 'second striker'), ('CF', 'center forward'), ('LF', 'left forward'), ('RF', 'right
forward')
ON CONFLICT (position_code) DO NOTHING;

CREATE TABLE IF NOT EXISTS awards (
    award_id      VARCHAR(5)    PRIMARY KEY CHECK (award_id ~ '^A-[0-9]+$'),
    award_name     VARCHAR(50)  NOT NULL UNIQUE,
    award_description VARCHAR(100) NOT NULL,
    year_introduced SMALLINT    NOT NULL CHECK (year_introduced BETWEEN 1930 AND 2100)
);
COMMENT ON TABLE awards IS 'One row per individual award type (Golden Ball, Golden Boot, ...).';

CREATE TABLE IF NOT EXISTS stadiums (
    stadium_id      VARCHAR(5)    PRIMARY KEY CHECK (stadium_id ~ '^S-[0-9]{3}$'),
    stadium_name     VARCHAR(100)  NOT NULL,
    city_name        VARCHAR(60)   NOT NULL,

```

```

country_name          VARCHAR(60) NOT NULL,
stadium_capacity      INTEGER      NOT NULL CHECK (stadium_capacity > 0),
stadium_wikipedia_link VARCHAR(255),
city_wikipedia_link   VARCHAR(255),
CONSTRAINT uq_stadium_name_city UNIQUE (stadium_name, city_name) -- two 'Olympiastadion's exist
(Berlin, Munich)
);
COMMENT ON TABLE stadiums IS 'One row per stadium that hosted a World Cup match.';

CREATE TABLE IF NOT EXISTS teams (
    team_id            VARCHAR(4)   PRIMARY KEY CHECK (team_id ~ '^T-[0-9]{2}$'),
    team_name          VARCHAR(60) NOT NULL UNIQUE,
    team_code          CHAR(3)      NOT NULL CHECK (team_code ~ '^[A-Z]{3}$'), -- NOT unique:
Germany and West Germany share DEU
    federation_name    VARCHAR(100) NOT NULL,
    region_name        VARCHAR(40) NOT NULL,
    confederation_id   VARCHAR(5)   NOT NULL REFERENCES confederations (confederation_id),
    team_wikipedia_link VARCHAR(255),
    federation_wikipedia_link VARCHAR(255)
);
COMMENT ON TABLE teams IS 'One row per national team, including historical teams (West Germany,
Soviet Union, ...). confederation_id is the current confederation.';

-- -----
-- 2. People
-- -----

CREATE TABLE IF NOT EXISTS players (
    player_id          VARCHAR(7)   PRIMARY KEY CHECK (player_id ~ '^P-[0-9]{5}$'),
    family_name         VARCHAR(60) NOT NULL,
    given_name         VARCHAR(60), -- NULL for players known by one name (e.g.
Pele)
    birth_date         DATE CHECK (birth_date >= DATE '1880-01-01'), -- NULL when unknown (78
players)
    player_wikipedia_link VARCHAR(255)
);
COMMENT ON TABLE players IS 'One row per player named in any World Cup squad.';

CREATE TABLE IF NOT EXISTS managers (
    manager_id         VARCHAR(5)   PRIMARY KEY CHECK (manager_id ~ '^M-[0-9]{3}$'),
    family_name         VARCHAR(60) NOT NULL,
    given_name         VARCHAR(60),
    country_name        VARCHAR(60) NOT NULL, -- nationality
    manager_wikipedia_link VARCHAR(255)
);
COMMENT ON TABLE managers IS 'One row per manager (head coach).';

CREATE TABLE IF NOT EXISTS referees (
    referee_id         VARCHAR(5)   PRIMARY KEY CHECK (referee_id ~ '^R-[0-9]{3}$'),
    family_name         VARCHAR(60) NOT NULL,
    given_name         VARCHAR(60),
    country_name        VARCHAR(60) NOT NULL, -- nationality
    confederation_id   VARCHAR(5)   NOT NULL REFERENCES confederations (confederation_id),
    referee_wikipedia_link VARCHAR(255)
);
COMMENT ON TABLE referees IS 'One row per referee appointed to a World Cup.';

-- -----

```

-- 3. Tournament structure

```
CREATE TABLE IF NOT EXISTS tournaments (  
    tournament_id VARCHAR(7) PRIMARY KEY CHECK (tournament_id ~ '^WC-[0-9]{4}$'),  
    tournament_name VARCHAR(40) NOT NULL UNIQUE,  
    year SMALLINT NOT NULL UNIQUE CHECK (year BETWEEN 1930 AND 2100),  
    start_date DATE NOT NULL,  
    end_date DATE NOT NULL,  
    CONSTRAINT ck_tournament_dates CHECK (end_date >= start_date),  
    CONSTRAINT ck_tournament_id_year CHECK (tournament_id = 'WC-' || CAST(year AS VARCHAR(4))),  
    CONSTRAINT ck_tournament_start_year CHECK (EXTRACT(YEAR FROM start_date) = year)  
);  
COMMENT ON TABLE tournaments IS 'One row per men's World Cup final tournament. Host, winner and  
format flags are derived (see views).';
```

```
CREATE TABLE IF NOT EXISTS tournament_stages (  
    tournament_id VARCHAR(7) NOT NULL REFERENCES tournaments (tournament_id),  
    stage_number SMALLINT NOT NULL CHECK (stage_number >= 1),  
    stage_name VARCHAR(20) NOT NULL CHECK (stage_name IN (  
        'group stage', 'second group stage', 'final round', 'round of 16',  
        'quarter-finals', 'semi-finals', 'third-place match', 'final')),  
    start_date DATE NOT NULL,  
    end_date DATE NOT NULL,  
    count_teams SMALLINT NOT NULL CHECK (count_teams > 0), -- kept: not derivable (1938  
walkover)  
    count_scheduled SMALLINT NOT NULL CHECK (count_scheduled >= 0),  
    count_playoffs SMALLINT NOT NULL CHECK (count_playoffs >= 0),  
    count_walkovers SMALLINT NOT NULL CHECK (count_walkovers >= 0),  
    PRIMARY KEY (tournament_id, stage_number),  
    CONSTRAINT uq_stage_name UNIQUE (tournament_id, stage_name),  
    CONSTRAINT ck_stage_dates CHECK (end_date >= start_date)  
);  
COMMENT ON TABLE tournament_stages IS 'One row per stage of a tournament. stage_name is stored only  
here.';
```

```
CREATE TABLE IF NOT EXISTS groups (  
    tournament_id VARCHAR(7) NOT NULL,  
    stage_number SMALLINT NOT NULL,  
    group_name VARCHAR(10) NOT NULL,  
    PRIMARY KEY (tournament_id, stage_number, group_name),  
    FOREIGN KEY (tournament_id, stage_number) REFERENCES tournament_stages (tournament_id,  
stage_number)  
);  
COMMENT ON TABLE groups IS 'One row per group within a group stage. Group names repeat across stages,  
so stage_number is part of the key.';
```

-- 4. Participation

```
CREATE TABLE IF NOT EXISTS qualified_teams (  
    tournament_id VARCHAR(7) NOT NULL REFERENCES tournaments (tournament_id),  
    team_id VARCHAR(4) NOT NULL REFERENCES teams (team_id),  
    performance VARCHAR(20) NOT NULL CHECK (performance IN (  
        'group stage', 'second group stage', 'final round', 'round of 16',  
        'quarter-finals', 'semi-finals', 'third-place match', 'final')),  
    PRIMARY KEY (tournament_id, team_id)
```

```

);
COMMENT ON TABLE qualified_teams IS 'Bridge tournaments M:N teams. performance = furthest stage
reached (final = winner or runner-up).';

CREATE TABLE IF NOT EXISTS host_countries (
    tournament_id VARCHAR(7) NOT NULL,
    team_id        VARCHAR(4) NOT NULL,
    PRIMARY KEY (tournament_id, team_id),
    FOREIGN KEY (tournament_id, team_id) REFERENCES qualified_teams (tournament_id, team_id)
);
COMMENT ON TABLE host_countries IS 'Host team(s) of each tournament (2002 has two). The host's
finish is in qualified_teams/tournament_standings.';

CREATE TABLE IF NOT EXISTS tournament_standings (
    tournament_id VARCHAR(7) NOT NULL,
    position       SMALLINT  NOT NULL CHECK (position BETWEEN 1 AND 4),
    team_id        VARCHAR(4) NOT NULL,
    PRIMARY KEY (tournament_id, position),
    CONSTRAINT uq_standing_team UNIQUE (tournament_id, team_id),
    FOREIGN KEY (tournament_id, team_id) REFERENCES qualified_teams (tournament_id, team_id)
);
COMMENT ON TABLE tournament_standings IS 'Final top-four positions of each tournament (1 =
champion).';

CREATE TABLE IF NOT EXISTS group_standings (
    tournament_id VARCHAR(7)  NOT NULL,
    stage_number   SMALLINT   NOT NULL,
    group_name     VARCHAR(10) NOT NULL,
    position       SMALLINT   NOT NULL CHECK (position >= 1),
    team_id        VARCHAR(4)  NOT NULL,
    wins           SMALLINT   NOT NULL CHECK (wins >= 0),
    draws         SMALLINT   NOT NULL CHECK (draws >= 0),
    losses        SMALLINT   NOT NULL CHECK (losses >= 0),
    goals_for      SMALLINT   NOT NULL CHECK (goals_for >= 0),
    goals_against SMALLINT   NOT NULL CHECK (goals_against >= 0),
    advanced       BOOLEAN    NOT NULL,
    PRIMARY KEY (tournament_id, stage_number, group_name, position),
    CONSTRAINT uq_group_team UNIQUE (tournament_id, stage_number, team_id),
    FOREIGN KEY (tournament_id, stage_number, group_name) REFERENCES groups (tournament_id,
stage_number, group_name),
    FOREIGN KEY (tournament_id, team_id) REFERENCES qualified_teams (tournament_id, team_id)
);
COMMENT ON TABLE group_standings IS 'Final group tables. played, goal difference and points are
derived (points rule changed in 1994).';

CREATE TABLE IF NOT EXISTS squads (
    tournament_id VARCHAR(7) NOT NULL,
    team_id        VARCHAR(4) NOT NULL,
    player_id      VARCHAR(7) NOT NULL REFERENCES players (player_id),
    shirt_number   SMALLINT  CHECK (shirt_number BETWEEN 1 AND 99),      -- NULL = unknown (source
used 0)
    position_code VARCHAR(3) NOT NULL REFERENCES positions (position_code)
                                CHECK (position_code IN ('GK', 'DF', 'MF', 'FW')),
    PRIMARY KEY (tournament_id, team_id, player_id),
    CONSTRAINT uq_squad_player UNIQUE (tournament_id, player_id),      -- a player is in one squad
per tournament
    CONSTRAINT uq_squad_shirt UNIQUE (tournament_id, team_id, shirt_number),
    FOREIGN KEY (tournament_id, team_id) REFERENCES qualified_teams (tournament_id, team_id)

```

```

);
COMMENT ON TABLE squads IS 'Bridge (tournament, team) M:N players. Event tables reference this
composite key.';

CREATE TABLE IF NOT EXISTS manager_appointments (
    tournament_id VARCHAR(7) NOT NULL,
    team_id        VARCHAR(4) NOT NULL,
    manager_id     VARCHAR(5) NOT NULL REFERENCES managers (manager_id),
    PRIMARY KEY (tournament_id, team_id, manager_id),          -- joint managers exist
    FOREIGN KEY (tournament_id, team_id) REFERENCES qualified_teams (tournament_id, team_id)
);
COMMENT ON TABLE manager_appointments IS 'Bridge (tournament, team) M:N managers.';

CREATE TABLE IF NOT EXISTS referee_appointments (
    tournament_id VARCHAR(7) NOT NULL REFERENCES tournaments (tournament_id),
    referee_id     VARCHAR(5) NOT NULL REFERENCES referees (referee_id),
    PRIMARY KEY (tournament_id, referee_id)
);
COMMENT ON TABLE referee_appointments IS 'Bridge tournaments M:N referees.';

-- -----
-- 5. Matches
-- -----

CREATE TABLE IF NOT EXISTS matches (
    match_id          VARCHAR(10) PRIMARY KEY CHECK (match_id ~ '^M-[0-9]{4}-[0-9]{2}$'),
    tournament_id     VARCHAR(7)  NOT NULL,
    stage_number      SMALLINT    NOT NULL,
    group_name        VARCHAR(10),                                -- NULL outside group
    stages
    replayed          BOOLEAN      NOT NULL,                      -- drawn and replayed
    later
    replay            BOOLEAN      NOT NULL,                      -- this match is the
    replay
    match_date        DATE         NOT NULL,
    match_time        TIME         NOT NULL,                      -- local kick-off time
    stadium_id        VARCHAR(5)  NOT NULL REFERENCES stadiums (stadium_id),
    home_team_id      VARCHAR(4)  NOT NULL,
    away_team_id      VARCHAR(4)  NOT NULL,
    home_team_score    SMALLINT    NOT NULL CHECK (home_team_score >= 0),
    away_team_score    SMALLINT    NOT NULL CHECK (away_team_score >= 0),
    extra_time        BOOLEAN      NOT NULL,
    penalty_shootout   BOOLEAN      NOT NULL,
    home_team_score_penalties SMALLINT CHECK (home_team_score_penalties >= 0), -- NULL when no
    shoot-out
    away_team_score_penalties SMALLINT CHECK (away_team_score_penalties >= 0),
    CONSTRAINT uq_match_tournament UNIQUE (match_id, tournament_id), -- target for composite FKs
    below
    CONSTRAINT ck_match_id_year CHECK (SUBSTRING(match_id FROM 3 FOR 4) = SUBSTRING(tournament_id
    FROM 4 FOR 4)),
    CONSTRAINT ck_match_teams_differ CHECK (home_team_id <> away_team_id),
    CONSTRAINT ck_match_replay CHECK (NOT (replayed AND replay)),
    CONSTRAINT ck_match_replayed_draw CHECK (NOT replayed OR home_team_score = away_team_score),
    CONSTRAINT ck_match_shootout CHECK (
        (penalty_shootout AND extra_time
         AND home_team_score = away_team_score
         AND home_team_score_penalties IS NOT NULL AND away_team_score_penalties IS NOT NULL
         AND home_team_score_penalties <> away_team_score_penalties)

```

```

        OR (NOT penalty_shootout AND home_team_score_penalties IS NULL AND away_team_score_penalties
IS NULL)),
    FOREIGN KEY (tournament_id, stage_number) REFERENCES tournament_stages (tournament_id,
stage_number),
    FOREIGN KEY (tournament_id, stage_number, group_name) REFERENCES groups (tournament_id,
stage_number, group_name),
    FOREIGN KEY (tournament_id, home_team_id) REFERENCES qualified_teams (tournament_id, team_id),
    FOREIGN KEY (tournament_id, away_team_id) REFERENCES qualified_teams (tournament_id, team_id)
);
COMMENT ON TABLE matches IS 'One row per match (replays are separate matches). Result, margins and
score text are derived.';

CREATE TABLE IF NOT EXISTS team_appearances (
    match_id          VARCHAR(10) NOT NULL,
    team_id           VARCHAR(4)  NOT NULL,
    tournament_id     VARCHAR(7)  NOT NULL,
    opponent_id       VARCHAR(4)  NOT NULL,
    home_team         BOOLEAN      NOT NULL,
    goals_for         SMALLINT     NOT NULL CHECK (goals_for >= 0),
    goals_against     SMALLINT     NOT NULL CHECK (goals_against >= 0),
    penalties_for     SMALLINT     CHECK (penalties_for >= 0),           -- NULL when no shoot-out
    penalties_against SMALLINT     CHECK (penalties_against >= 0),
    PRIMARY KEY (match_id, team_id),
    CONSTRAINT ck_ta_opponent CHECK (team_id <> opponent_id),
    CONSTRAINT ck_ta_penalties CHECK ((penalties_for IS NULL) = (penalties_against IS NULL)),
    FOREIGN KEY (match_id, tournament_id) REFERENCES matches (match_id, tournament_id),
    FOREIGN KEY (tournament_id, team_id) REFERENCES qualified_teams (tournament_id, team_id),
    -- the opponent must have its own row for the same match; checked at COMMIT so both rows can be
inserted together
    CONSTRAINT fk_ta_opponent FOREIGN KEY (match_id, opponent_id) REFERENCES team_appearances
(match_id, team_id)
    DEFERRABLE INITIALLY DEFERRED
);
COMMENT ON TABLE team_appearances IS 'Each match seen from one team (2 rows per match). Kept as a
team-centred view of matches for simpler queries.';

CREATE TABLE IF NOT EXISTS referee_appearances (
    match_id          VARCHAR(10) PRIMARY KEY,           -- one referee per match
    tournament_id     VARCHAR(7)  NOT NULL,
    referee_id        VARCHAR(5)  NOT NULL,
    FOREIGN KEY (match_id, tournament_id) REFERENCES matches (match_id, tournament_id),
    FOREIGN KEY (tournament_id, referee_id) REFERENCES referee_appointments (tournament_id,
referee_id)
);
COMMENT ON TABLE referee_appearances IS 'The referee of each match (1:1 with matches).';

CREATE TABLE IF NOT EXISTS manager_appearances (
    match_id          VARCHAR(10) NOT NULL,
    team_id           VARCHAR(4)  NOT NULL,
    manager_id        VARCHAR(5)  NOT NULL,
    tournament_id     VARCHAR(7)  NOT NULL,
    PRIMARY KEY (match_id, team_id, manager_id),
    FOREIGN KEY (match_id, tournament_id) REFERENCES matches (match_id, tournament_id),
    FOREIGN KEY (match_id, team_id) REFERENCES team_appearances (match_id, team_id),
    FOREIGN KEY (tournament_id, team_id, manager_id) REFERENCES manager_appointments (tournament_id,
team_id, manager_id)
);
COMMENT ON TABLE manager_appearances IS 'Manager(s) in charge of a team in a match.';

```

```

-----
-- 6. Match events
-- tournament_id is kept in event tables on purpose: it lets the composite FK to squads check that
-- the player was in that team's squad for that tournament. The FK (match_id, tournament_id) ->
matches
-- guarantees it always agrees with the match, so the extra column cannot become inconsistent.
-----

CREATE TABLE IF NOT EXISTS player_appearances (
    match_id      VARCHAR(10) NOT NULL,
    player_id     VARCHAR(7)  NOT NULL,
    tournament_id VARCHAR(7)  NOT NULL,
    team_id       VARCHAR(4)  NOT NULL,
    position_code VARCHAR(3)  NOT NULL REFERENCES positions (position_code),
    starter       BOOLEAN     NOT NULL,                -- FALSE = came on as a
substitute
    captain       BOOLEAN     NOT NULL,
    PRIMARY KEY (match_id, player_id),
    FOREIGN KEY (match_id, tournament_id) REFERENCES matches (match_id, tournament_id),
    FOREIGN KEY (match_id, team_id) REFERENCES team_appearances (match_id, team_id),
    FOREIGN KEY (tournament_id, team_id, player_id) REFERENCES squads (tournament_id, team_id,
player_id)
);
COMMENT ON TABLE player_appearances IS 'Line-ups from 1970: one row per player who played in a
match.';

CREATE TABLE IF NOT EXISTS goals (
    goal_id       VARCHAR(6)  PRIMARY KEY CHECK (goal_id ~ '^G-[0-9]{4}$'),
    match_id      VARCHAR(10) NOT NULL,
    tournament_id VARCHAR(7)  NOT NULL,
    team_id       VARCHAR(4)  NOT NULL,                -- team credited with the
goal
    player_id     VARCHAR(7)  NOT NULL,
    player_team_id VARCHAR(4)  NOT NULL,                -- scorer's own team
    minute_regulation SMALLINT NOT NULL CHECK (minute_regulation BETWEEN 1 AND 120),
    minute_stoppage SMALLINT  NOT NULL DEFAULT 0 CHECK (minute_stoppage >= 0),
    own_goal      BOOLEAN     NOT NULL,
    penalty       BOOLEAN     NOT NULL,                -- in-match penalty (not
shoot-out)
    CONSTRAINT ck_goal_own_goal CHECK (own_goal = (team_id <> player_team_id)),
    CONSTRAINT ck_goal_not_both CHECK (NOT (own_goal AND penalty)),
    FOREIGN KEY (match_id, tournament_id) REFERENCES matches (match_id, tournament_id),
    FOREIGN KEY (match_id, team_id) REFERENCES team_appearances (match_id, team_id),
    FOREIGN KEY (match_id, player_team_id) REFERENCES team_appearances (match_id, team_id),
    FOREIGN KEY (tournament_id, player_team_id, player_id) REFERENCES squads (tournament_id, team_id,
player_id)
);
COMMENT ON TABLE goals IS 'One row per goal. Own goals: credited to team_id, scored by a player of
player_team_id.';

CREATE TABLE IF NOT EXISTS bookings (
    booking_id    VARCHAR(6)  PRIMARY KEY CHECK (booking_id ~ '^B-[0-9]{4}$'),
    match_id      VARCHAR(10) NOT NULL,
    tournament_id VARCHAR(7)  NOT NULL,
    team_id       VARCHAR(4)  NOT NULL,
    player_id     VARCHAR(7)  NOT NULL,
    minute_regulation SMALLINT NOT NULL CHECK (minute_regulation BETWEEN 1 AND 120),

```

```

minute_stoppage    SMALLINT    NOT NULL DEFAULT 0 CHECK (minute_stoppage >= 0),
yellow_card        BOOLEAN     NOT NULL,
red_card           BOOLEAN     NOT NULL,                                -- straight red
second_yellow_card BOOLEAN     NOT NULL,                                -- second yellow -> sent
off
CONSTRAINT ck_booking_some_card CHECK (yellow_card OR red_card),
CONSTRAINT ck_booking_second_yellow CHECK (NOT second_yellow_card OR yellow_card),
FOREIGN KEY (match_id, tournament_id) REFERENCES matches (match_id, tournament_id),
FOREIGN KEY (match_id, team_id) REFERENCES team_appearances (match_id, team_id),
FOREIGN KEY (tournament_id, team_id, player_id) REFERENCES squads (tournament_id, team_id,
player_id)
);
COMMENT ON TABLE bookings IS 'Cards from 1970. Sent off = red_card OR second_yellow_card.';

CREATE TABLE IF NOT EXISTS substitutions (
    substitution_id VARCHAR(6) PRIMARY KEY CHECK (substitution_id ~ '^S-[0-9]{4}$'),
    match_id        VARCHAR(10) NOT NULL,
    tournament_id   VARCHAR(7)  NOT NULL,
    team_id         VARCHAR(4)  NOT NULL,
    player_id       VARCHAR(7)  NOT NULL,
    minute_regulation SMALLINT NOT NULL CHECK (minute_regulation BETWEEN 1 AND 120),
    minute_stoppage SMALLINT NOT NULL DEFAULT 0 CHECK (minute_stoppage >= 0),
    coming_on       BOOLEAN     NOT NULL,                                -- FALSE = going off
    FOREIGN KEY (match_id, tournament_id) REFERENCES matches (match_id, tournament_id),
    FOREIGN KEY (match_id, team_id) REFERENCES team_appearances (match_id, team_id),
    FOREIGN KEY (tournament_id, team_id, player_id) REFERENCES squads (tournament_id, team_id,
player_id)
);
COMMENT ON TABLE substitutions IS 'Substitutions from 1970: two rows per substitution (one going off,
one coming on).';

CREATE TABLE IF NOT EXISTS penalty_kicks (
    penalty_kick_id VARCHAR(6) PRIMARY KEY CHECK (penalty_kick_id ~ '^PK-[0-9]{3}$'),
    match_id        VARCHAR(10) NOT NULL,
    tournament_id   VARCHAR(7)  NOT NULL,
    team_id         VARCHAR(4)  NOT NULL,
    player_id       VARCHAR(7)  NOT NULL,
    converted       BOOLEAN     NOT NULL,
    FOREIGN KEY (match_id, tournament_id) REFERENCES matches (match_id, tournament_id),
    FOREIGN KEY (match_id, team_id) REFERENCES team_appearances (match_id, team_id),
    FOREIGN KEY (tournament_id, team_id, player_id) REFERENCES squads (tournament_id, team_id,
player_id)
);
COMMENT ON TABLE penalty_kicks IS 'Penalty shoot-out kicks (1982+). In-match penalties are in goals
(penalty = TRUE).';

-- -----
-- 7. Awards
-- -----

CREATE TABLE IF NOT EXISTS award_winners (
    tournament_id VARCHAR(7) NOT NULL,
    award_id      VARCHAR(5) NOT NULL REFERENCES awards (award_id),
    player_id     VARCHAR(7) NOT NULL,
    team_id       VARCHAR(4) NOT NULL,
    PRIMARY KEY (tournament_id, award_id, player_id),                                -- ties give several winners
    FOREIGN KEY (tournament_id, team_id, player_id) REFERENCES squads (tournament_id, team_id,
player_id)

```

```

);
COMMENT ON TABLE award_winners IS 'Award winners per tournament. Shared = more than one winner for
the same award and tournament.';

-- -----
-- 8. Indexes
-- Primary keys and UNIQUE constraints are indexed automatically. These extra indexes cover
-- foreign-key columns that are not the leading column of an existing index and that the
-- analytical queries join or filter on.
-- -----

CREATE INDEX IF NOT EXISTS ix_teams_confederation          ON teams (confederation_id);
CREATE INDEX IF NOT EXISTS ix_referees_confederation      ON referees (confederation_id);
CREATE INDEX IF NOT EXISTS ix_qualified_teams_team        ON qualified_teams (team_id);
CREATE INDEX IF NOT EXISTS ix_squads_player              ON squads (player_id);
CREATE INDEX IF NOT EXISTS ix_manager_appointments_mgr    ON manager_appointments (manager_id);
CREATE INDEX IF NOT EXISTS ix_referee_appointments_ref    ON referee_appointments (referee_id);
CREATE INDEX IF NOT EXISTS ix_matches_stage              ON matches (tournament_id, stage_number);
CREATE INDEX IF NOT EXISTS ix_matches_stadium            ON matches (stadium_id);
CREATE INDEX IF NOT EXISTS ix_matches_home_team          ON matches (home_team_id);
CREATE INDEX IF NOT EXISTS ix_matches_away_team          ON matches (away_team_id);
CREATE INDEX IF NOT EXISTS ix_team_appearances_team      ON team_appearances (team_id, opponent_id);
CREATE INDEX IF NOT EXISTS ix_referee_appearances_ref    ON referee_appearances (referee_id);
CREATE INDEX IF NOT EXISTS ix_manager_appearances_mgr    ON manager_appearances (manager_id);
CREATE INDEX IF NOT EXISTS ix_player_appearances_player  ON player_appearances (player_id);
CREATE INDEX IF NOT EXISTS ix_goals_match_team          ON goals (match_id, team_id);
CREATE INDEX IF NOT EXISTS ix_goals_player              ON goals (player_id);
CREATE INDEX IF NOT EXISTS ix_bookings_match            ON bookings (match_id, team_id);
CREATE INDEX IF NOT EXISTS ix_bookings_player           ON bookings (player_id);
CREATE INDEX IF NOT EXISTS ix_substitutions_match        ON substitutions (match_id, team_id);
CREATE INDEX IF NOT EXISTS ix_penalty_kicks_match       ON penalty_kicks (match_id, team_id);
CREATE INDEX IF NOT EXISTS ix_award_winners_player      ON award_winners (player_id);

-- End of DDL: 28 tables (27 from the dataset + positions lookup), 21 secondary indexes.

```

Data: Joshua C. Fjelstul, *The Fjelstul World Cup Database*, <https://github.com/jfjelstul/worldcup>, CC-BY-SA 4.0.
Modifications: tables normalised (columns removed as listed in Section 3.2), text markers converted to NULL, one lookup table added, and the group names of 18 matches (1982 second stage, 1950 final round) aligned with groups.
No data values were changed. This document is a derived work under the same licence.